Markets

Shuo Liu

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Introduction

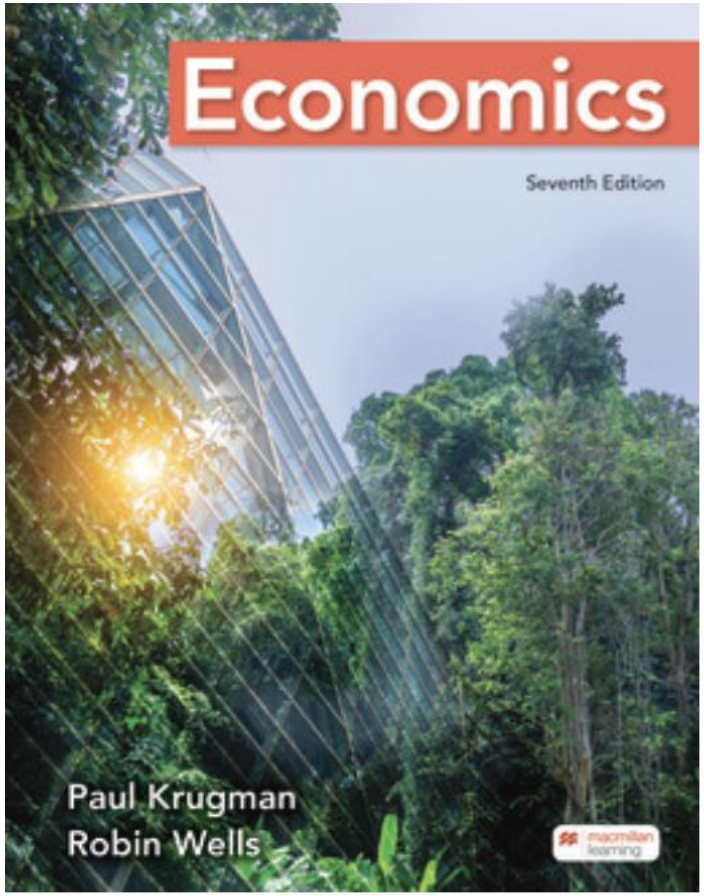
- Shuo Liu (刘烁), PhD in Economics, University of Zurich
- Visiting Associate Professor of Economics, NYU Abu Dhabi
- Associate Professor of Economics, Peking University
- <u>Research interests:</u>
  - Game theory
  - Information economics
  - Industrial and organizational economics
  - Behavioral economics
- <u>Teaching</u>: *Game Theory* (UG, MBA), *Organizational Economics* (UG, PhD), *Microeconomics* (UG, PhD) , *Topics in Behavioral/Political Economics* (PhD)

Course Information

- **Lecturer:** Shuo Liu
  - *Email: sl4015@nyu.edu*
  - *Office: A5-181*
  - *Office hours: Friday 2:00-3:30 pm (email first) , or by appointment*

- **Teaching Assistant:** Amit Goyal
  - *Email: ag10131@nyu.edu*
  - *Office hours: By appointment*

Textbook



Krugman & Wells

Microeconomics, 7th edition

(E-book access available)

Course Information

- Slides: posted to Brightspace before lecture.
- Assignments: weekly, via Brightspace and/or Achieve.
- Recitations: see Albert or Brightspace for schedules.
- Midterms: in class, TBD
- Final: see Albert for schedule.
- Evaluation:

Participation: 10%	Two midterms: 20% each
Assignments: 15%	Final: 35%

Week	Topic	Chapters
1	First Principles; Economic Models	1, 2
2	Supply and Demand	3
3	Consumer and Producer Surplus	4
4	Price Controls and Quotas; Elasticity	5, 6
5	Taxes	7
	Midterm I	
6	International Trade	8
7	Decision Making	9
8	The Rational Consumer	10
9	Behind the Supply Curve	11
10	Perfect Competition	12
	Midterm II	
11	Monopoly	13
12	Oligopoly, Monopolistic Competition	14, 15
13	Externalities	16
14	Public Goods, Revision	17
	Final	

Classroom Polices

- You are expected to arrive on time and stay for the full class. But if you must be late or leave early, please do so quietly.
- Be respectful and open-minded during discussions.
- Feel free to interrupt me with questions at any time.
- Silence your phone before class begins.
- Please avoid any vertical-screen devices (e.g., laptops) during lecture. Paper and pen are encouraged for note-taking.

AI Usage

- You are encouraged to complete assignments without relying on AI.
- We will not attempt to detect AI use.
- Consider what dependence on AI means for your exam performance.
- Consider what it means for your own learning and skill development.

The AI Becker problem

Who will train the next generation?

LUIS GARICANO
JUL 23, 2025

32 10 9 Share

To pay individuals like $100 million over a four-year package, that's actually pretty cheap compared to the value created for the business. — Benjamin Mann, Anthropic Cofounder

Do we need armies of business analysts creating PowerPoints? No, the technology could do that — Kate Smaje, Head of technology and AI, McKinsey and Co.

When Anthropic's cofounder defends nine-figure salaries for AI researchers, he inadvertently highlights a paradox. While Meta and Google engage in bidding wars for senior AI talent — with packages reaching $100 million — the same technology is eliminating the entry-level positions that once created such expertise. Paraphrasing McKinsey head of tech's statement to Bloomberg above, what is the point of junior analysts?

What Is Economics?

[E]conomics is the social science that studies the production, distribution, and consumption of goods and services.

— Krugman & Wells

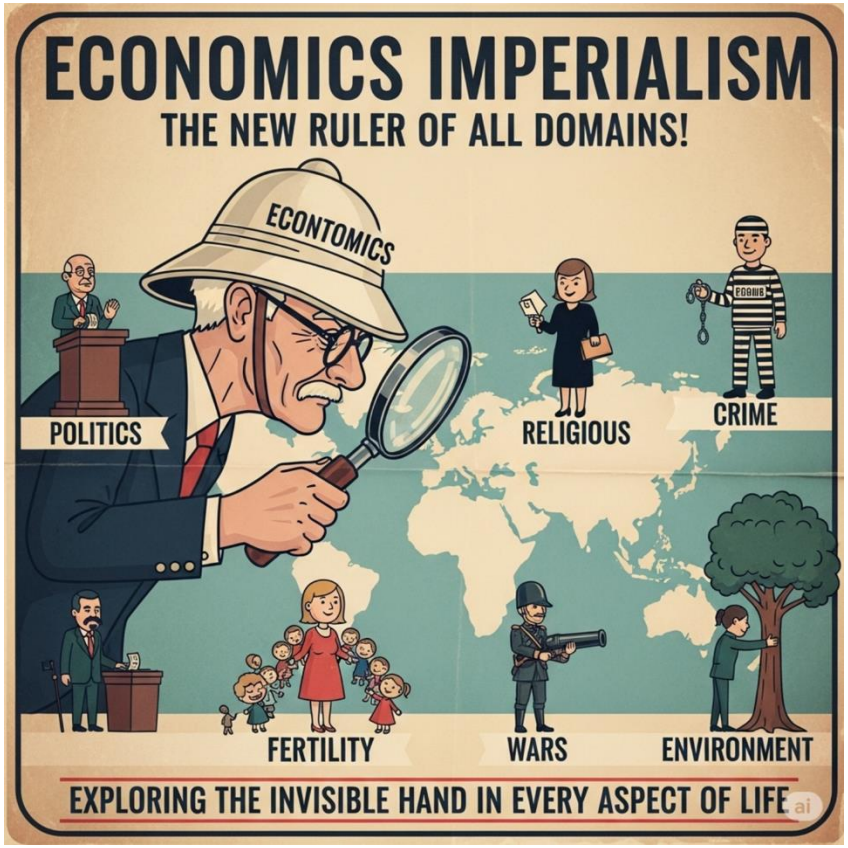
Economics is a study of mankind in the ordinary business of life.

— Alfred Marshall

Economics can be defined in a few different ways. It's the study of scarcity, the study of how people use resources and respond to incentives, or the study of decision-making. It often involves topics like wealth and finance, but it's not all about money.

— The American Economic Association

What Is Economics?



Gary Becker | The first economic imperialist

Gary Becker was an economist with rare imagination. He extended the domain of economic rationality to explain complex social behaviours

Siddharth Singh
Published • 6 May 2014, 06:51 PM IST

Add mint

Photo:Bloomberg

Why Study Economics?

The purpose of studying economics is not to acquire a set of ready-made answers to economic questions, but to learn how to avoid being deceived by economists.

— Joan Robinson

Studying economics can't guarantee you won't be unemployed, but when you find yourself in the unemployment line one day, the economics you've learned will help you understand why you're there.

— Some economist

Why Study Economics?

Median Annual Wages for Economists
(In the top 5 industries employing economists)

Industry	Annual Wage
Federal government, excluding postal service	$141,590
Scientific research and development services	$129,430
Management, scientific, and technical consulting services	$102,450
Local government, excluding education and hospitals	$92,440
State and local government, excluding education and hospitals	$74,520

Source: Bureau of Labor Statistics, U.S. Department of Labor, Occupational Outlook Handbook, Economists. (Visited June 5, 2025).

Chapter 1: First Principles

First Principles

- 4 principles guiding how individuals make choices
- 4 principles governing how individual choices interact
- 3 principles illustrating how the economy as a whole works

1. People Must Make Choices Because Resources Are Scarce

- Choices: What's feasible ⟹ What matters most
- Resources: Money, time, attention, manpower,…
- Making choices is about how these resources are allocated among competing uses:
  - Food or clothing?
  - Leisure or work?
  - Instagram accounts to follow?
  - Task assignments among staff?
- Choices are necessary because resources are limited.

2. The True Cost of Something Is Its Opportunity Cost

- The opportunity cost of a choice is what you must give up to get it.
- If multiple things are given up, the opportunity cost is the one with the highest value to you.
- In this sense, nothing is truly free: the money, time or energy spent on one thing could always be used elsewhere.

Is the true cost of coming to college same for everyone?

Mark Zuckerberg.
Credit : Mark Zuckerberg/Instagram

Broken Window Theory or Fallacy?

- Story:

A child breaks a barber shop's window → Demand for glass rises → Workers get jobs and wages

- Reasoning:

$$\text{Demand} \Longrightarrow \text{Consumption} \Longrightarrow \text{Production} \Longrightarrow \text{Economic growth!}$$

- Conclusion 1: The barber's misfortune is society's gain.
- Conclusion 2: Destruction improves social welfare.
- A theory or a fallacy?

3. "How Much" Decisions Require Trade-Offs at the Margin

- Marginal changes are small, incremental adjustments to what you're already doing.
- Good decisions come from comparing marginal benefits with marginal costs.

One more slice of pizza?

One more hour of study?

4. People (Usually) Respond to Incentives

- People react when marginal benefits or costs change.
- If the marginal benefit of an action exceeds its marginal cost, they will choose it.

Case study: A US-based installer of automobile glass (Lazear, 2000)

Pre-1994: Hourly wage → After-1994: Piece rate (pay for every window installed)

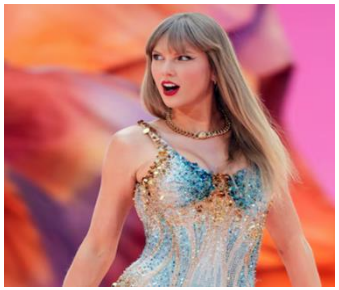
- Pay to workers? 10% ↑
- Productivity? 44% ↑
- Profits and customer satisfaction ↑
- Sick leave ↓

- Motivational effect only accounted for roughly 20% of the productivity gain.
- Can you guess what drove the rest?

5. There Are Gains From Trade

- Trade occurs when people exchange goods or services.
- In markets, trade happen only when both sides expect to gain.

- The main source of gains from trade is specialization—when people focus on what they do best.



6. Markets Move Toward Equilibrium

- Equilibrium: a situation in which no individuals can be better off by doing something different ↔ no unexploited opportunities.
- Why equilibrium? Because people respond to incentives! (Principle 4)

What would you think if you saw an empty checkout lane?

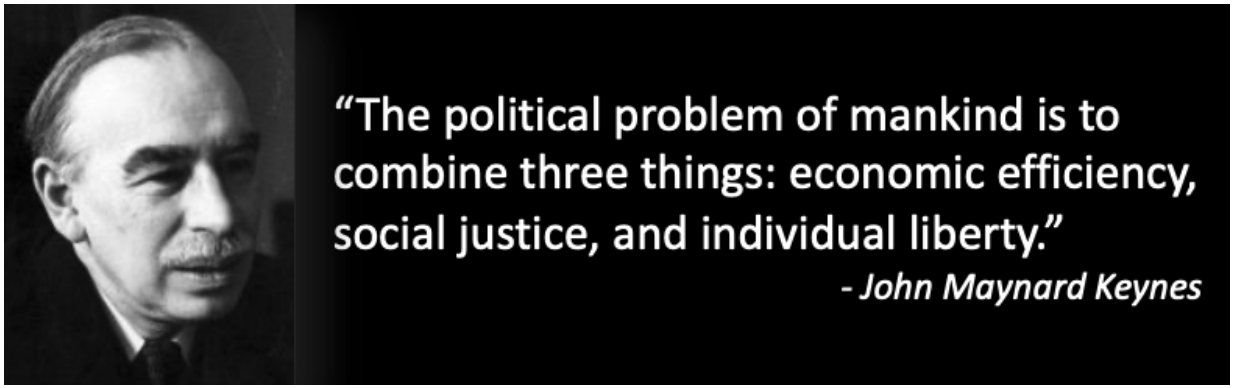
- In reality, the market may not be *exactly* at the equilibrium.
- However, equilibrium forces are always at play and provide powerful insights into how markets work.

7. Resources Should Be Used as Efficiently as Possible

- We all have an intuitive sense of what inefficiency looks like in daily life:

- Sometimes equity (or fairness) trumps efficiency.

7. Resources Should Be Used as Efficiently as Possible



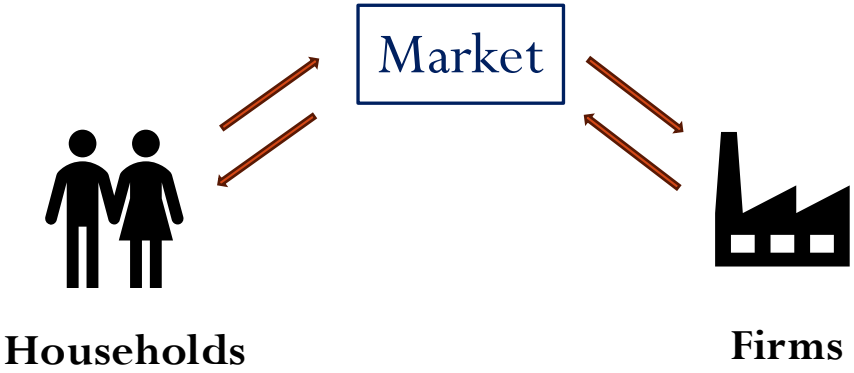
- Social justice: everyone gets their "fair" share
- Economic efficiency: producing things with the least possible resources
- Individual liberty: people are free to choose for themselves

Pareto Efficiency: No opportunities remain to make someone better off without making someone else worse off.

8. Markets Usually Lead to Efficiency, …

"It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest. We address ourselves, not to their humanity but to their self-love, and never talk to them of our own necessities but of their advantages."

— Adam Smith, *The Wealth of Nations*



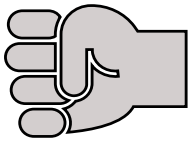
8. (continued) but if They Don't, Government Intervention Can Improve Society's Welfare

Market Power

Externality

Information Asymmetry

Market Failure

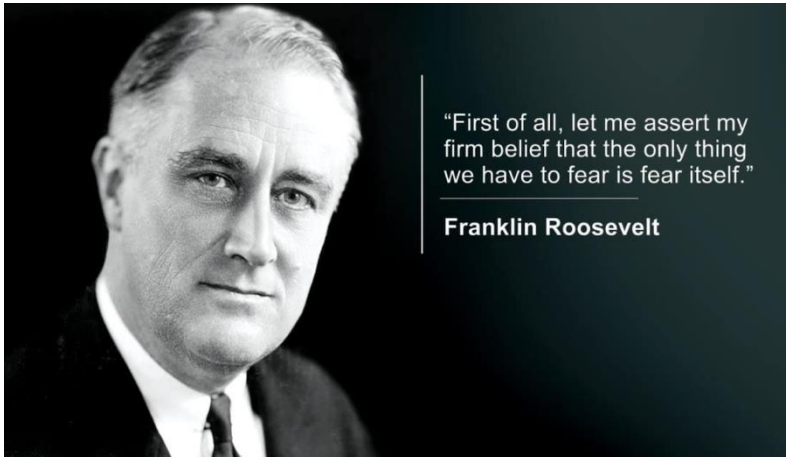


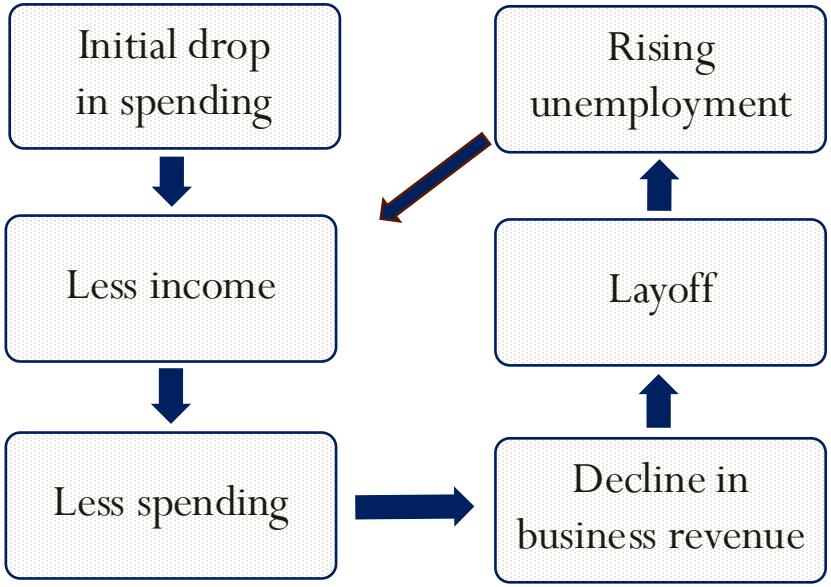
The Visible Hand

9. One Person's Spending Is Another Person's Income

- Chain reaction through all these 'Spending = Income' interactions

Recession

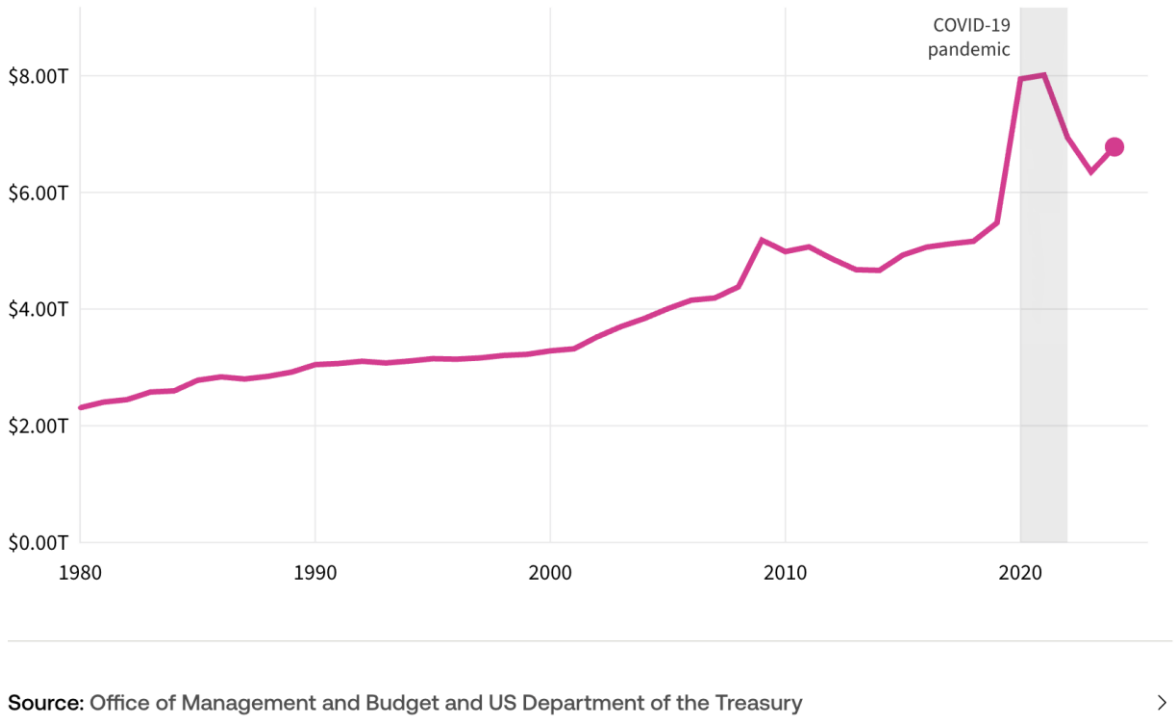




A similar reinforcing loop emerges during economic booms…

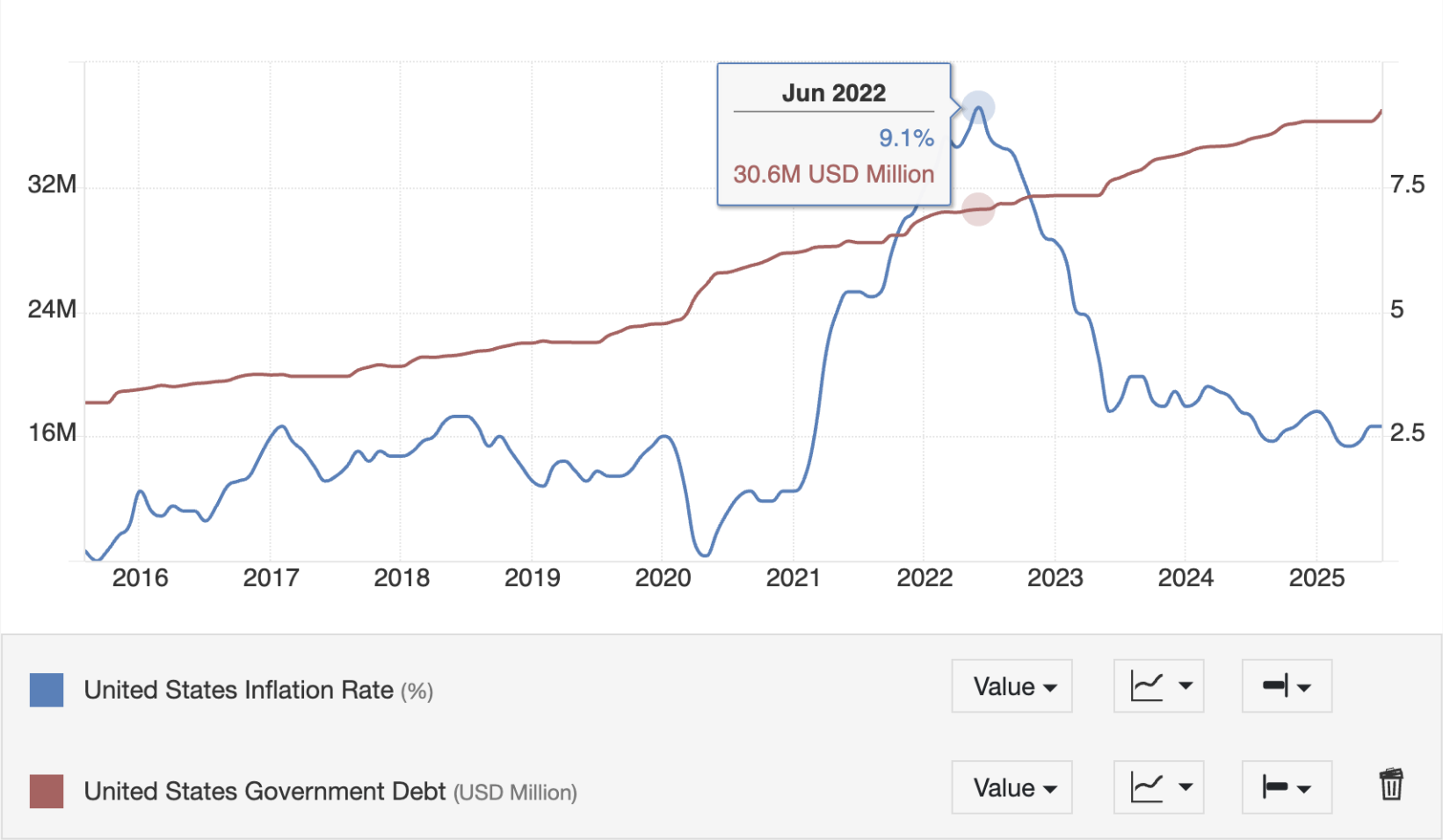
10. Spending Sometimes Gets out of Line with Productive Capacity; When It Does, Government Policies Can Change Spending

- Government policies can be used to address the imbalances between the economy's spending and supply.



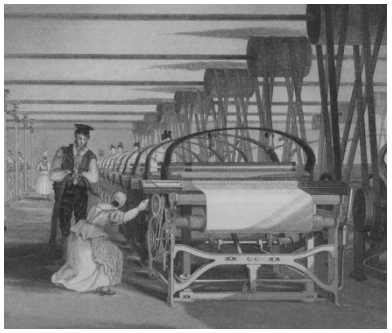
Federal government spending, adjusted for inflation (FY 2024 USD)

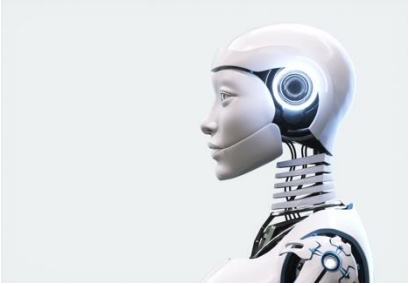
10. Spending Sometimes Gets out of Line with Productive Capacity; When It Does, Government Policies Can Change Spending



11. Increase in The Economy Potential Leads to Economic Growth Over Time

Country	Avg TFP (1995–2004)	Avg TFP (2005–2014)	Avg TFP (2015–2024)	GDP/Cap 1995 (PPP)	GDP/Cap 2024 (PPP)	Rank Shift
USA	1.40%	0.60%	0.80%	$28,700	$85,810	Stable
UK	1.20%	0.20%	0.30%	$21,100	$58,900	Down (~15)
Germany	0.80%	0.60%	0.10%	$22,400	$72,300	Down (~10)
Japan	0.40%	0.40%	0.20%	$23,400	$51,685	Down (~30)
China	3.50%	3.00%	1.20%	$1,400	$27,105	Up (~45)
India	1.50%	2.50%	2.00%	$1,700	$11,160	Up (~15)
Brazil	0.20%	0.30%	-0.20%	$8,200	$22,338	Down (~10)
Argentina	0.50%	0.30%	-0.80%	$11,000	$30,431	Down (~15)
Ethiopia	-0.50%	2.50%	3.00%	$580	$3,287	Up (~10)
Rwanda	0.50%	2.50%	3.00%	$550	$3,475	Up (~12)





Summary

- Introduction
- Chapter 1: Eleven core principles
- Next lecture, Chapter 2: Economic Models